3. Social Media Text, Action & Hyperlink analytics

Q.1 What is text analytics, and explain the steps in text analytics with an example.

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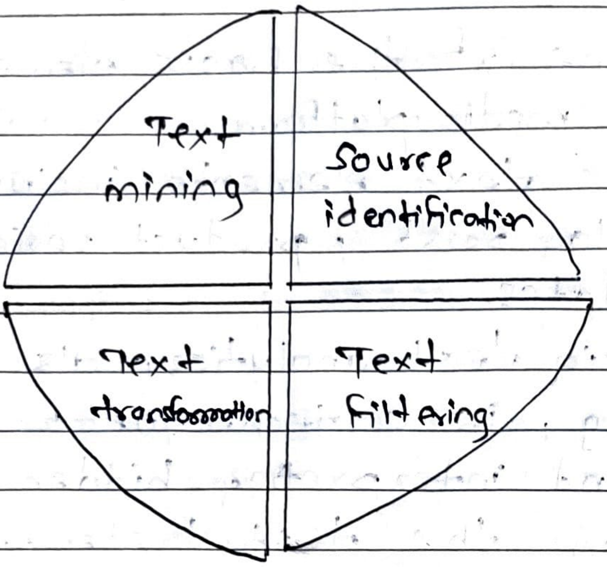
- Text is one of the basic elements of any social media platform.
- social media text elements include comments, tweets, blog post, product reviews and status updates.
- Social media text analytics, also known as text mining, is a technique for extracting, analysing and interpreting hidden business insights from the textual elements of social media content.
- Organizations use text analytics techniques to extract valuable hidden meaning, patterns, and structures from user-generated social media text.
- Types of text:

1) Dynamic text
=) Dynamic text is text or statements that social media users generate in real-time to express their opinions about content or information posted via social media.
- Dynamic text is typically shorter, more varied in nature and updated or deleted more frequently.

2) Static text
=) Static social media text tends to be very long and is unlikely to be created, updated or deleted.

- Steps :



1) Identification and searching

⇒ The text analysis process begins by identifying the source of the text to be analysed.

- Texts posted on social media are dynamic, large, diverse, multilingual and noisy.
- Finding the right source of text analytics is therefore very important for gaining useful business insights.
- The genre of the source text also determines the kind of tools used to extract and analyse it.

2) Text parsing and filtering

⇒ The next step is to use NPL, which is primarily based on ML techniques to analyse, clean and filter the text and create a dictionary of words.

- In order for computers and algorithms to extract meaning from text, sentence structure and parts of speech are determined, named entities are extracted, stopwords are removed and spelling is checked.
- Most of these steps are automatic. However, certain stages require human intervention.
- For e.g., the filtering stage may require manual cleanup to remove unwanted or irrelevant terms.

3) Text transformation

⇒ To apply an analysis algorithm to text, it must be transformed into a machine-readable format for analysis.
- The cleaned text is therefore computed using linear-algebra based techniques such as : B latent semantic and vector space model converted to numerical representation.

4) Text Mining

⇒ This step actually breaks down the text to extract the business insights you need.
- Various text mining algorithm are applied to the text.
- Clustering, association, classification, predictive analytics, sentiment analysis.
- Text analytics uses these sophisticated algorithms to extract sentiment.

Social Media Text Analytics tools:

1) Discovertext
⇒ It is a powerful platform for collecting, cleaning and analyzing text and social media data streams.

2) Lexalytics
⇒ Lexalytics is a social media text and semantic analysis tool for social media platform, including Twitter, Facebook, blogs.

3) Tweet Archivist
⇒ Tweet Archiwist is focused on searching archiving, analyzing and visualizing tweets based on a search term or hashtags.

4) Voyant
⇒ Voyant is a web-based text reading and analysis.
- With voyant body of the text can be read from a file or directly exported from website.

5) Netlytics
⇒ Netlytics is a cloud-based text and social media analytics platform for social media text that discovers social networks from online conversations on social media.

Q.2 Difference between static and dynamic text.

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Static text	Dynamic text
1) Text that does not change once created.	1) Text that changes based on context or user interaction.
2) Sources are books, articles, brochures, manuals.	2) Sources are websites, social media feeds, chatbots or user dashboards.
3) No interaction - the context is fixed.	3) Interactive - the content may change with user input.
4) Rarely updated or manually revised.	4) Frequently updated, often automatically.
5) Sutaible for formal, archieved or reference info.	5) Sutaible for real-time communication or personalization.
6) For e.g., text in PDF document, printed books.	6) For e.g., ~~Facebook~~ text in Facebook post, twitter timelines.

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Q. 3 Note on : Intension analysis in social media.

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- Intent or Intent Mining extracts user intent from social media natural language text such as user comments, product reviews, tweets intended to investigate.
- Unlike sentiment analysis, intention analysis tries to understand what the users wants to do.
- The deeper insight helps:

i) Brands respond to customer needs.
ii) Marketers target users based on behavior.
iii) Organizations detect early signs of crises.
iv) Chatbots understand user goals more effectively.

- Steps

- 1) Data collection
- 2) Text preprocessing
- 3) Feature extraction
- 4) Intention classification
- 5) Intention categories
- 6) Post-processing & Action

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1) Data collection
⇒ Collect text data from platform like Twitter, Facebook, Instagram, Reddit.
- Example: Tweets mentioning a product or service.

2) Text preprocessing
⇒ Cleans the data: remove stopwords, punctuation, hashtag, URLs.
- Tokenize and lemmatize words to standardize them.
- Example:
Raw: "I wanna buy an iphone! Any suggestions?"
Cleaned: "want buy iphone suggestions"

3) Feature extraction
⇒ Convert text into numerical features using methods like:

a) TF-IDF
b) Word embeddings
c) POS tags and dependency parsing.

4) Intent classification
⇒ Use machine learning or deep learning models to classify intent:

a) ML models: Logistic regression, SVM, Naive Bayes.
b) NLP Models: BERT, RoBERTa, XLNet.

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5) Intention Categories
⇒ Intent vary by use-cases can include:

a) Purchase Intent : "Looking to buy new headphones."
b) Complaint : "Worst delivery experience ever".
c) Feedback : "Great app! Very smooth experience".
d) Appreciation : "Thanks for the fast response!"

6) Post-processing & Action
⇒ Filters users based on intent and take action:

a) Recommend products to high-intent buyers.
b) Assign complaints to customer service
c) Flag misinformation or abusive content.

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Q.4 Explain social media action analytics, Common social media actions and Action Analytics tools.

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- Actions are the ways of expressing symbolic reactions to the social media content.
- Actions are not just symbolic reactions. They have emotions and actions available.
- Typical actions taken by social media users include likes, dislikes, shares, views, clicks, tags, etc.
- Analysing social media actions deals with extracting, analysing and interpreting the insights contained in the actions of social media users.
- Businesses can use action analytics to measure the popularity and impact of their products, services or ideas through social media.
- Common actions :
  - 1) Likes
  - 2) Dislikes
  - 3) Tagging
  - 4) Mentions
  - 5) View
  - 6) Clicks
  - 7) Uploading and Downloading

1) Like

⇒ The like button is a feature on social media sites that allows users to express their positive feelings towords certain products, services, people, etc.

- The like button is often incorporated into social media platforms and websites and the no. of likes received by content is often displayed.

2) Dislikes

⇒ The Dislike button on social media platforms such as YouTube allows users to express their negative feeling towards certain content.

- This feature is visible to others and the no. of dislikes is accumalated over time.

3) Tagging

⇒ Tagging is the act of adding extra information to social media content for identification and classification purposes.

- It can take various forms, such as attaching descriptive keywords to posts or adding tags to photos and comments on Facebook.

- Tagging fastens the process of searching and finding relevant content.

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4) Mentions

⇒ Mentions or social mentions refer to the occurence of a person, place, or thing being mentioned by name on social media.
- A twitter mention is the inclusion of a @username in a tweet.

5) Views

⇒ Views are the number of times social media content is viewed by users.
- Page views are the no. of times a page on a company website or blog is viewed by a visitor.
- View data can be used to understand user engagement and populosity of contend.

6) Clicks

⇒ Clicks are actions performed by users when they click on hyperlinked content on a website or blog.
- Click data can be used for business intelligence purposes, such as improving website traffic and reducing bounce rate.

7) Uploading and Downloading

⇒ Uploading is the act of adding new content to a social media platform.
- The opposite of uploading is downloading that is, the act of receiving data from a social media platform.

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• Action Analytics tools

1) Hootsuite

⇒ Manages social media presence across popular networks, offers various plans including free, pro and enterprise version.

2) Lithium

⇒ Provides a range of products and services for social media ~~engagement~~ management, including analytics, marketing and crowd-sourcing.

3) Google Analytics

⇒ Tracks and analyzes website traffic can also be used for blog and wiki analytics.

4) Facebook Insights

⇒ Facebook insights help Facebook page owners understand and analyze user growth and demographic trends.

5) Hashtagify

⇒ This tool measures the influence of hashtags.

Q.5 Explain Hyperlink analytics? Explain tools of hyperlink analytics.

=>

- Hyperlinks are paths of social media traffic.
- A hyperlink is a reference to a webpage that users can click to access.
- Types of Hyperlinks :
  - 1) Inlink
  - 2) Outlink
  - 3) Colink

1) Inlinks

=> Inlinks are hyperlinks or links pointing to one website or from another website.

- For e.g.,

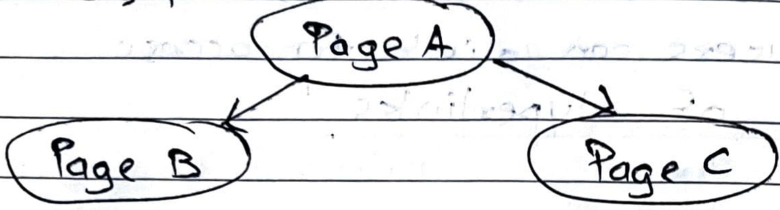
- Page A gets two Inlinks from page B and Page C.
- Inlinks are very interesting for social markers as they drive traffic to a particular website.
- Inlink also play an important role in website analytics, as both the quality and no. of inlinks can affect a website's ~~reach~~ search engine rankings.
- Inlinks also affect the popularity of your social media content.

2) Out-links

⇒ Out-links are hyperlinks generated from within a website.

- These links provide additional resources, references or citation.

- For e.g.,



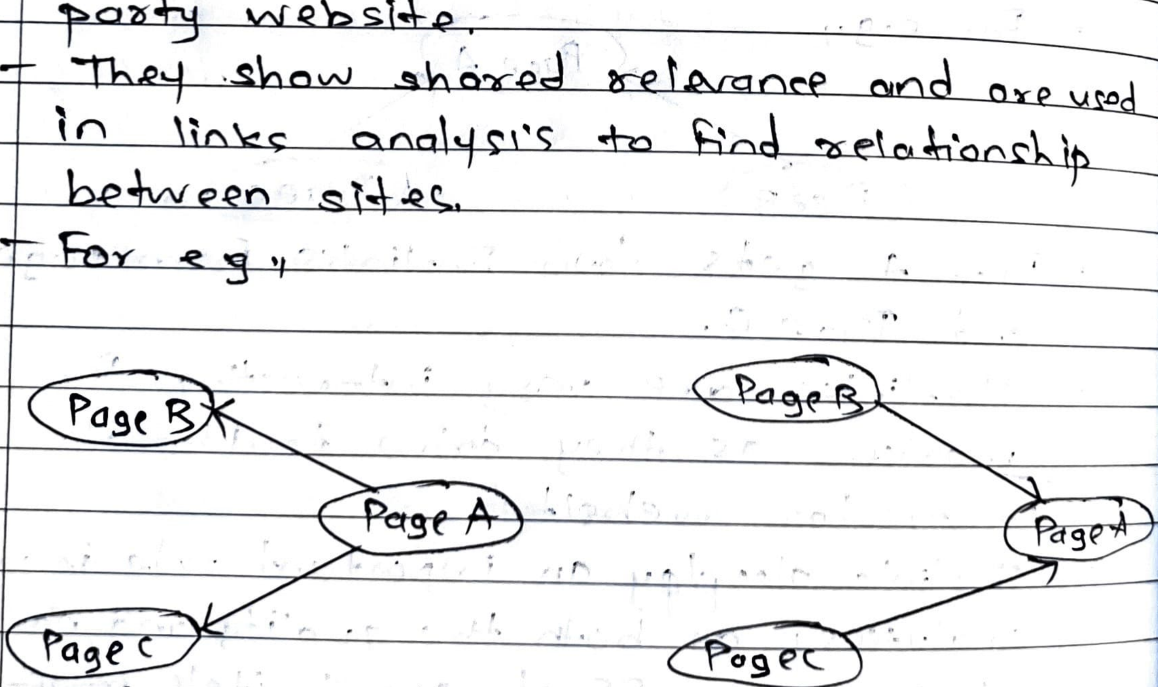
Page A sending to outlines to page B and Page C.

3) Co-links

⇒ Co-links occur when two different websites both links to the same third-party website.

- They show shared relevance and are used in links analysis to find relationship between sites.

- For e.g.,



Page A links to the both pages B and C so B and C are considered to be interlinked or indirectly connected.

- Hyperlink analysis deals with the extraction analysis and interpretation of hyperlinks.
- A basic premise of hyperlink analysis is that the number and quality of hyperlinks to a website reflect their importance or value.
- Types of Hyperlink Analytics :-
  - 1) Hyperlink Environment Analysis
  - 2) Link Impact Analysis
  - 3) Social media hyperlink analysis.

1) Hyperlink Environment Analysis

⇒ Hyperlink proximity analysis looks at a specific website or group of websites.

- Extract and analyse website hyperlinks to identify sources of internet traffic.
- There are two form of hyperlink boundary networks :-

i) Co-link Network : Nodes are sites and links represent the similarity between the sites as measured by no. of co-links.

ii) Inlink & Out-link Network : A hyperlink neighbourhood network is built on the inlinks and outlinks from a site or set of sites. In such network, nodes are websites and links represent inlinks and outlinks.

2) Link Impact Analysis

⇒ A link impact analysis examines the impact of a website address on the web in terms

of citation and mentions across the web.
- Link impact analysis collects and analyze statistics about web pages that mention a particular website URL.

3) Social Media Hyperlink Analysis
⇒ Social media hyperlink analysis deals with the extraction and analysis of hyperlinks embedded in social media text.
- By extracting and examining these hyperlinks you can identify the source and destination of your social media traffic.

• Hyperlink Analytics Tools

1) Webometric Analyst
⇒ Webometric analyst is a web impact analysis tool that allows you to perform a variety of analyses on social media platform, including hyperlink network analysis and web mentions.

2) VOSON
⇒ VOSON is a hyperlink analysis tool for building and analysing hyperlink networks.

3) Link Diagnosis
⇒ Link diagnosis is a free online tool for analyzing and diagnosing links.

4) Majestic

⇒ Majestic offers a variety of link analysis tools, including a link explorer, backlink history and link mapping tools.

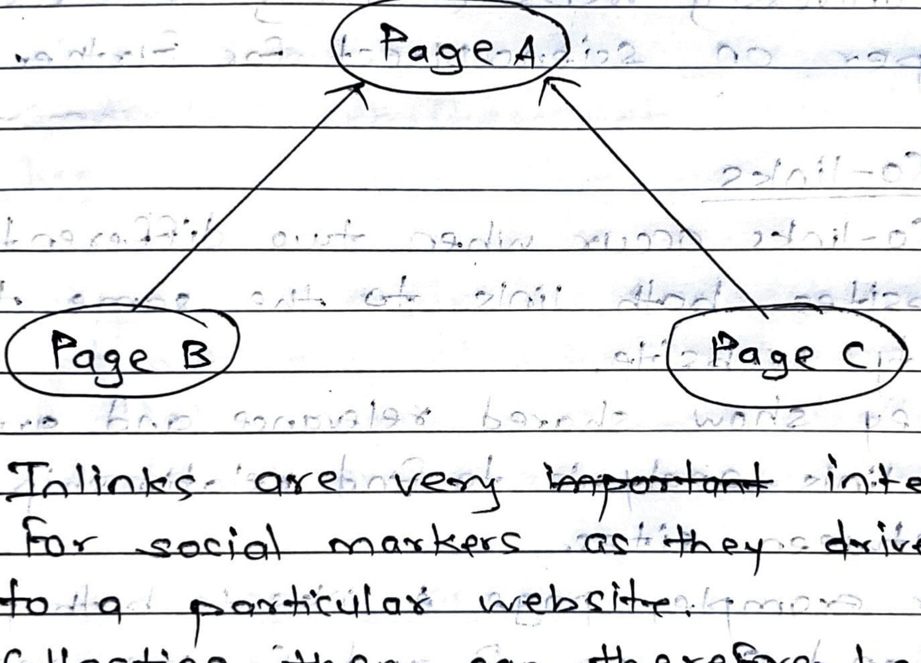
5) Backlink Watch

⇒ Backlink monitoring is a free tool for checking the quality and quantity of inlinks to your website.

Q. Briefly discuss in-links, out-links and co-links.

1) **Inlinks**

⇒ In-links are hyperlinks from external website pointing to another website.
- These links helps increase a site's authority, credibility and SEO ranking.
- For e.g., page A gets two In-links from page B and page C as shown below -



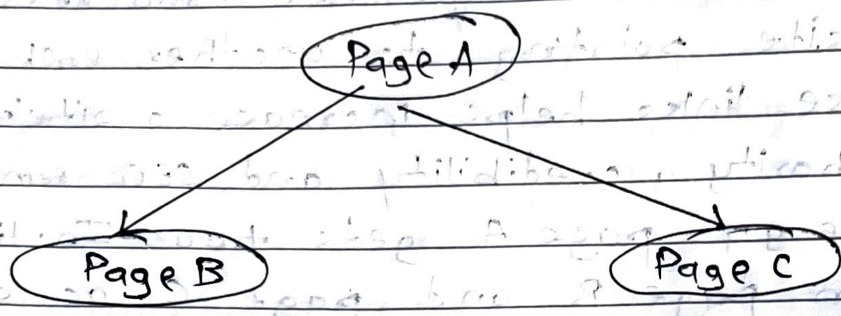
- Inlinks are very ~~important~~ interesting for social markers as they drive traffic to a particular website.
- Collecting them can therefore help you understand where the traffic to your business website is coming from.
- For e.g., A tech blog linking to an article on Apple's official website about the latest iPhone.

2) **Out-links**

⇒ Out-links are hyperlinks from your website pointing to an external website.
- These links provide additional resources,

references or citation.

- For example, page A sending two out-links to page B and page C.

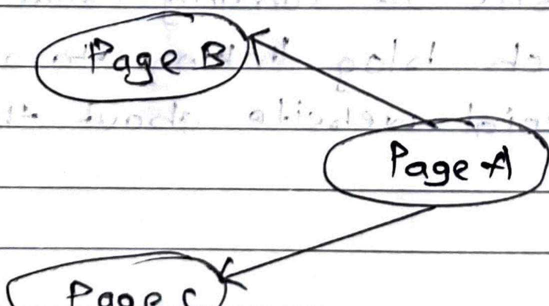


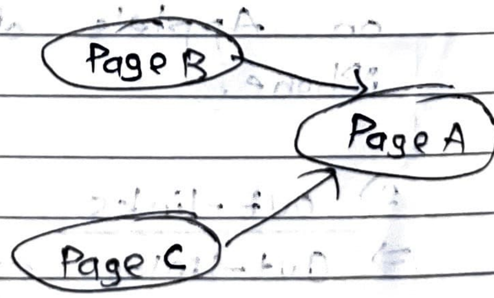
- For e.g.,
  A university website linking to a research paper on ScienceDirect for further reading.

3) Co-links

⇒ Co-links occur when two different websites both link to the same third-party website.

- They show shared relevance and are used in link analysis to find relationship between sites.
- For example, page A links to both pages B and C, so B and C are considered to be interlinked or indirectly connected.





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Q.14 Explain the four main purpose of social media text analytics.

⇒

1) Sentiment Analysis

⇒ Sentiment analysis involves categorizing social media text as positive, negative or neutral.

- It is often used to understand how customers feels about a product, service or issue.
- Tools like Semantria use algorithms to identify sentiment-bearing phrases in text and assign them a score based on a logarithmic scale.
- Sentiment analysis can provide valuable insights into the emotions and opinions of social media users.

2) Intention Mining

⇒ Intention mining involves discovering user's intentions from natural language social media text.

- Companies can use intention mining to identify potential customers and service existing customers who have issues with a product.
- Examples of intention-bearing phrases include "buy", "purchase" and "quit".
- Tools like Semantria can be used to mine intentions from social media text.

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3) Trend Mining

⇒ Trend mining, also known as predictive analytics, uses large amounts of historical and real-time social media data to predict future events.

- It involves identifying patterns and trends in social media data to improve products, services or customer satisfaction.
- Techniques used in trend mining include machine learning, data mining and social network analysis.

4) Concept Mining

⇒ Concept mining is a method for extracting ideas and concepts from documents.

- It is used to classify, cluster and rank these ideas.
- Concept mining is different from text mining, which focuses on extracting specific information rather than broader ideas and concepts.
- Examples of documents that can be analyzed using concept mining include social media text, web pages and news transcripts.

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Q.18 Why it important to measure actions performed by social media users?

⇒

- Measuring user actions on social media is crucial for businesses, marketers and organizations to understand audience behavior, improve engagement and optimize stratergies.

1) Understanding Audience Behavior

⇒ Analyzing likes, shares, comments and clicks helps businesses understand what content resonates with their audience.

- For e.g.,
A brand see's that video posts get 3X more engagement than text posts, so they shift their content stratergy towards videos.

2) Improving content stratergy

⇒ Tracking interactions help identify which types of posts perform best and refine content stratergies.

- For e.g.,
A company notices that "how-to-guide" get more shares than promotional posts, so they create more educational content.

3) Measuring marketing ROI

⇒ Businesses need to track how social media campaigns lead to sales, sign-ups, or website visits to measure success.

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- For e.g.,
A clothing brand runs a paid Instagram ad and tracks how many users clicked the "Shop Now" button to measure ad effectiveness.

4) Enhancing customer support
⇒ Tracking complaints and negative comments in real-time helps brands respond quickly and prevent PR disasters.
- For e.g.,
An airline detects a surge in complaints about flight delays on Twitter and immediately updates customer with solutions

5) Personalizing user experiences
⇒ Social media algorithms use user interactions to recommend personalized content and improve user experience.
- For e.g.,
Netflix tracks what users watch and suggests similar show's to keep them engaged.

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Q.19 What are hyperlinks and why they are important?

⇒

- A hyperlink is a clickable element in a digital document, webpage or application that ~~users~~ directs users to another direction such as web page, file, image, etc.
- Hyperlinks are usually embedded in text, images or buttons and are activated by clicking on them.

1) Enable Easy Navigation

⇒ Hyperlinks allows users to jump from one webpage to another quickly.
- They create a network of connected web pages, making the internet easy to explore.
- For e.g.,
Clicking on a Wikipedia reference takes you to related articles.

2) Improve User Experience

⇒ Well-placed hyperlinks make website more interactive, informative and user-friendly.
- Users can access additional content without manually searching for it.
- For e.g.,
A blog post about healthy diets ~~als~~ links to a detailed article on nutrition tips.

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3) SEO

⇒ Hyperlinks helps search engine like Google understand website structure.

- Websites with high-quality backlinks rank higher in search results.

- For e.g., A new webcite linking to authoritative sources improves credibility and SEO ranking.

4) Cross-Referencing

⇒ Academic and professional documents use hyperlinks to cite sources and reference related studies.

- For e.g.,
A research paper include hyperlinks to journal articles and datasets.

5) Helps in Online marketing

⇒ Businesses use hyperlinks in emails, ads and social media posts to direct users to product pages.

- For e.g.,
"Get 50% OFF!! Click here to show now!"

link to an e-commerce website.